PCA

Summary sheet — Deep Learning (S.-A. Berrani, ENSIA 2026–2027)

Core idea & notation

PCA is a purely *descriptive*, exploratory method (no probabilistic model assumed, normality not required) that handles any number of quantitative variables. It moves from p variables to $q < p$ new variables while distorting reality as little as possible.

1. Goal and principle

Goal: find the projection that **minimizes information loss**, i.e. the one that **maximizes the dispersion** of the projected data.
Variance (dispersion): mean squared distance between the points and their centroid c:

$$V = \frac{1}{n}\sum_{i=1}^{n} d^2(x_i, c)$$

Property: going from p to $q < p$ necessarily loses dispersion. The idea is to lose as little as possible by choosing the projection axes well.

Key principle of PCA. Choose the projection (the axes) so as to *maximize the dispersion* of the projected points.

2. Algorithm

Given a data matrix X of size $n \times p$ (n individuals, p quantitative variables):

1. **Center (and scale) data X (Normalization):**
   - Compute column means: $\bar{x}_k = \frac{1}{n}\sum_{i=1}^{n} x_{ik}$.
   - Center: $x_{ik} \leftarrow x_{ik} - \bar{x}_k$.
   - If variables have different units/scales, divide by standard deviation: $x_{ik} \leftarrow \frac{x_{ik}-\bar{x}_k}{\sigma_k}$.
2. **Calculate the covariance matrix:**
   $$C = X^T X \quad (p \times p)$$
3. **Compute eigenvalues and eigenvectors of $X^T X$:**
   - **Eigenvalues (a_i):** Solve the characteristic equation:
     $$\det(X^T X - aI) = 0$$
   - **Eigenvectors (v_i):** For each eigenvalue a_i, solve the linear system:
     $$(X^T X - a_i I)v_i = 0 \quad \text{such that} \quad \|v_i\| = 1 \quad \left(\sum_{k=1}^{p}(v_{ik})^2 = 1\right)$$
4. **Order the pairs (a_i, v_i) in descending order of eigenvalues:**
   $$a_1 \geq a_2 \geq \cdots \geq a_p \geq 0$$
   Here, the eigenvalue a_i equals the inertia (variance) carried by the principal axis v_i.
5. **Keep the first q pairs to construct the projection matrix V:**
   $$V = \begin{bmatrix} v_1 & v_2 & \dots & v_q \end{bmatrix} \quad (p \times q)$$
6. **Project the individuals into the new space:**
   $$U = XV \quad (n \times q) \qquad \text{entry by entry: } u_i^j = \langle x_i, v_j \rangle = \sum_{k=1}^{p} x_{ik} v_{jk}$$

	Centering only	Centering + scaling
Operation	subtract the mean of each variable	also divide by the standard deviation
Use when	variables are *homogeneous* (same unit, scale)	variables are *heterogeneous*

Outliers. PCA is very sensitive to noise: a few outliers can artificially rotate principal axes and alter the number of components needed to reach the target explained variance.

3. Worked example with X, U, V (toy data, $n = 4, p = 2$)

Setup: four individuals, two variables, columns already centered (sums = 0) and of the same scale:

$$X = \begin{pmatrix} 2 & 1 \\ 1 & 2 \\ -1 & -2 \\ -2 & -1 \end{pmatrix} \quad (4 \times 2)$$

1. **Center/reduce:** column means are $0 \Rightarrow$ nothing to subtract; same scale $\Rightarrow$ no scaling.
2. **Covariance matrix:** $X^T X = \begin{pmatrix} 10 & 8 \\ 8 & 10 \end{pmatrix}$.
3. **Eigen-decomposition:**

$$\det(X^T X - aI) = (10 - a)^2 - 64 = 0 \implies a_1 = 18, \quad a_2 = 2$$

Eigenvectors: $(X^T X - 18I)v_1 = 0 \implies v_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$; $\quad (X^T X - 2I)v_2 = 0 \implies v_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$.

4. **Sort:** $(a_1, v_1) = \left(18, \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}\right)$, then $(a_2, v_2) = \left(2, \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}\right)$.
5. **Keep $q = 1$:** $V = v_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ $(p \times q = 2 \times 1)$.
6. **Project:** $U = XV$ (4×1); with all axes kept, $V = [v_1 \ v_2]$ and U is 4×2:

$$U = XV = \frac{1}{\sqrt{2}} \begin{pmatrix} 3 & 1 \\ 3 & -1 \\ -3 & 1 \\ -3 & -1 \end{pmatrix}, \qquad \text{e.g. } u_1^1 = \frac{2 \cdot 1 + 1 \cdot 1}{\sqrt{2}} = \frac{3}{\sqrt{2}} \approx 2.12$$

Result: the first column of U $(= u^1)$ is the $q = 1$ representation. Reconstruction with $q = 1$: $X' = u^1 v_1^T$ gives row $1 = (1.5, 1.5)$ instead of $(2, 1)$; error $J = \|X - X'\|_F^2 = 4 \times 0.5 = 2 = a_2$ (the discarded eigenvalue). Explained variance: $\frac{18}{18+2} = \mathbf{90}\%$.

4. Explained variance: choosing q

Choose the smallest q such that the cumulative explained variance meets a chosen threshold s:

$$\frac{\sum_{i=1}^{q} a_i}{\sum_{i=1}^{p} a_i} \geq s$$

Since a_i is the inertia of axis i, this represents the proportion of total information retained.

Axis	1	2	3	4
Eigenvalue a_i (lecture, 4 grades)	28.23	12.03	0.03	0.01
Explained variance	0.70	0.30	0.00	0.00
Cumulative	0.70	1.00	1.00	1.00
Toy example above	18	2	—	—
Explained / cumulative	0.90	0.10 / 1.00	—	—

In the lecture's grades example ($\sum a_i = 40.30$), $q = 2$ retains $\approx 100\%$ of the total information.

5. Studying the variables: flip the table and use correlations

Idea: position yourself in a different space by transposing the data perspective. Each initial variable X^i (column i of X) and each principal component c^j (column j of U) is viewed as a vector in $\mathbb{R}^n$.

Goal: analyze the linear correlation coefficients $r(c^j, X^i)$ between initial variables and principal components.

1. **Compute principal components:** $U = XV$ (columns $c^j = u^j$).
2. **Compute correlation for every variable i and axis j:**

$$r(c^j, X^i) = \frac{\sum_k u_k^j x_k^i}{\sqrt{\sum_k (u_k^j)^2}\sqrt{\sum_k (x_k^i)^2}} \qquad \text{(since columns are centered)}$$

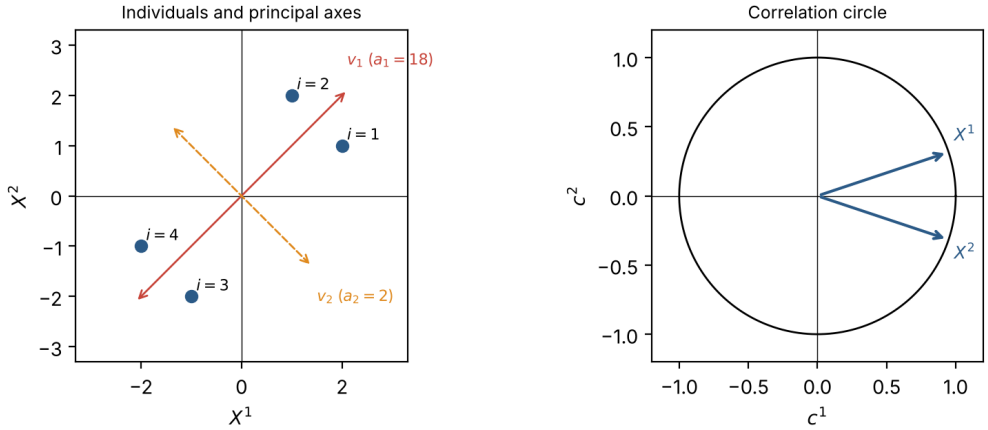
3. **Gather in a $p \times q$ table of correlations** ("variables-factors").
4. **Plot variable X^i as point $\big(r(c^1, X^i), r(c^2, X^i)\big)$ inside the correlation circle.**
5. **Interpretation:** arrows close to each other ⇒ strong positive correlation (≈ 1); orthogonal arrows ⇒ uncorrelated (≈ 0); opposite arrows ⇒ negative correlation (≈ -1).

Example (toy data): $x^1 = (2, 1, -1, -2)$, $x^2 = (1, 2, -2, -1)$, $u^1 = \frac{1}{\sqrt{2}}(3, 3, -3, -3)$, $u^2 = \frac{1}{\sqrt{2}}(1, -1, 1, -1)$:

$$r(c^1, X^1) = \frac{18/\sqrt{2}}{\sqrt{18}\sqrt{10}} = \frac{3}{\sqrt{10}} \approx 0.95, \qquad r(c^2, X^1) = \frac{\sqrt{2}}{\sqrt{2}\sqrt{10}} = \frac{1}{\sqrt{10}} \approx 0.32$$

Variable	c^1 (90%)	c^2 (10%)
X^1	0.95	0.32
X^2	0.95	−0.32

Shortcut: since $X^T u^j = a_j v_j$, $r(c^j, X^i) = v_j^i \frac{\sqrt{a_j}}{\|x^i\|}$. Check: $0.95^2 + 0.32^2 = 1$ (length = 1 on the circle).



Left: individuals and principal axes v_1, v_2. Right: correlation circle, X^1 and X^2 are strongly correlated.

Reading the circle. Long arrows close to the circumference are well-represented by the retained axes; short arrows near the center are poorly represented and should not be interpreted.